

When does an emoji’s fit with the sentence start to matter?

Late sentence–emoji congruency in Chinese sarcasm reading: eye-tracking evidence from a fully crossed sentence type × emoji valence design

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Readers work out whether a face emoji fits the sentence once they look at the emoji, and then they go back to the word.

Background

- Congruent/incongruent object emojis affect the word before them during the **first fixation** on that word (Sheridan et al., 2024). Face emojis were excluded.
- In Chinese digital communication the WeChat smiley 😊 has become a conventional **sarcasm marker** (Huang et al., 2025), and it contradicts the negative intentions when being sarcastic, while congruent to the semantic meaning.
- And face emojis carry **evaluative and pragmatic** information, where object emojis carry **referential** information.
- Whether that kind of information is available parafoveally and how pragmatic incongruency takes place remain untested in sarcastic emoji scenarios.

Design

Participants. 65 native Mandarin speakers, mean age 23.4.

Design. Pre-registered 2 (Sentence Type) × 3 (Emoji Valence) within-subjects, with 48 experimental items and 44 fillers over six counterbalanced lists. The emoji is sentence-final, so it sits in the parafovea while the evaluative word is fixated.

Analysis. Mixed-effects models on reading times, with equivalence tests and Bayes factors for effects close to zero.

MEASURES: WHAT THE EYES WERE SCORED ON

Two regions: (1) The **evaluative word**, read while the emoji is still in the parafovea, so effects there reflect **preview**; (2) and **the emoji** itself, where effects reflect what happens once the eye lands on it.

EARLY: THE FIRST VISIT

First fixation duration. The first fixation landing in the region.

Gaze duration. All fixations before the eye leaves the region.

First-run fixation count. How many fixations that first visit contains.

Skipping. The region is passed over on the first pass.

LATE: RE-READING AND LOOKS BACK

Go-past time. From first entering the region until the eye moves past it, including looks back to earlier text.

Total time. Every fixation in the region, first pass and re-reading together.

Regressions out. Trials with a leftward move out of the region.

Regressions back to the word. Trials with a later look back into the evaluative word.

Pre-registered prediction: congruent pairings shorten first-pass reading of the evaluative word and raise its skipping rate.

Materials

DOES THE EMOJI FIT THE WORD?

	smiley 😊	frowny 😞
Sarcastic positive word	fits	clashes
Literal negative word	clashes	fits

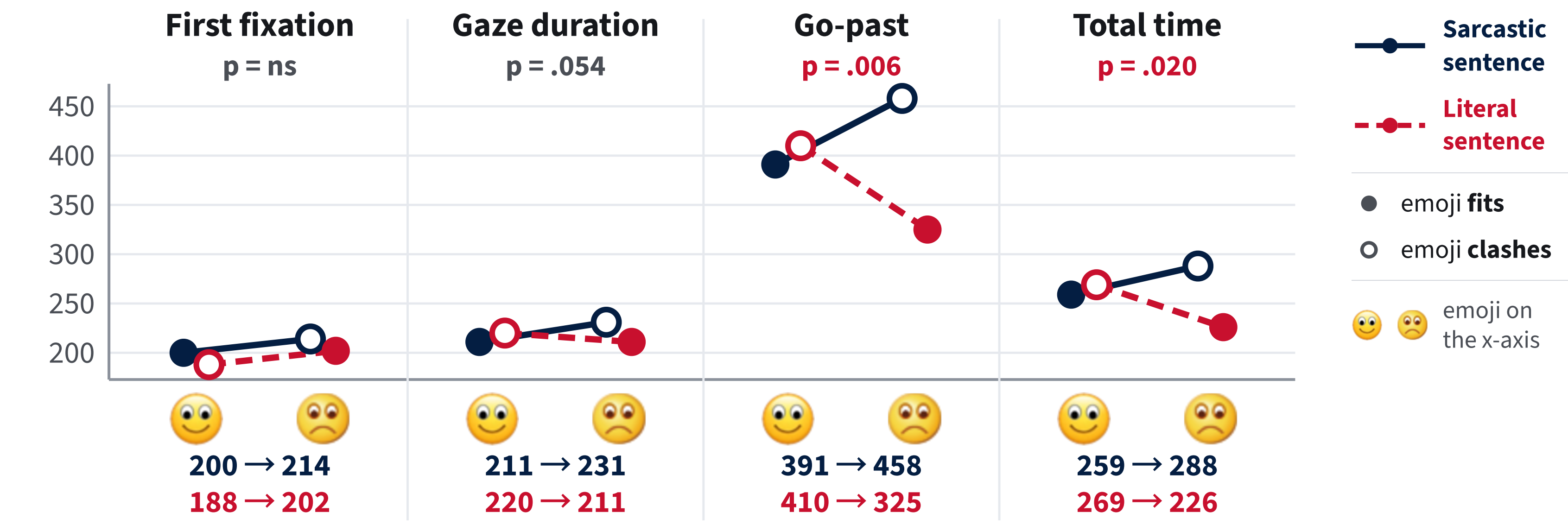
A no-emoji version of every item serves as the baseline in both sentence types.

Sentence	Emoji	Example	Match
Sarcastic	none	王璐发现你的画不是很好看。她发消息给你：画得 太好了 。 [Wang Lu found your drawing is quite bad. She messaged you: What a beautiful drawing]	baseline
Sarcastic	smiley 😊	...画得 太好了 😊 [...What a beautiful drawing 😊]	Congruent
Sarcastic	frowny 😞	...画得 太好了 😞 [...What a beautiful drawing 😞]	Incongruent
Literal	none	王璐发现你的画不是很好看。她发消息给你：画得 太差了 。 [Wang Lu found your drawing is quite bad. She messaged you: What a bad drawing]	baseline
Literal	smiley 😊	...画得 太差了 😊 [...What a bad drawing 😊]	Incongruent
Literal	frowny 😞	...画得 太差了 😞 [...What a bad drawing 😞]	Congruent
Filler	monkey 🐵	李明要去图书馆还书。他发消息给你：路上有点堵 🐵 [Li Ming is returning books. He messaged you: Traffic is a bit heavy 🐵] Neutral WeChat monkey faces, colour matched to the experimental emojis.	not analysed
Comprehension question	after every trial	王璐觉得你的画好看吗? [Did Wang Lu think your drawing was good?] Accuracy 95.5%.	yes / no

Boxed characters are the critical region, with the evaluative word in red; the emoji is sentence-final, so it sits in the parafovea while the critical word is fixated. **48 experimental items and 44 fillers** over six counterbalanced lists; items adapted from Huang et al. (2025).

1 · The cost of a clashing emoji shows up at the emoji, in five measures

READING TIME ON THE EMOJI, BY-PARTICIPANT MEANS (MS), ONE SHARED SCALE



The crossover grows over time: the two lines run parallel on the first fixation and cross once later measures are reached. Overall reading times for the smiley and the frowny were equal on every measure, so each face is equally easy on its own.

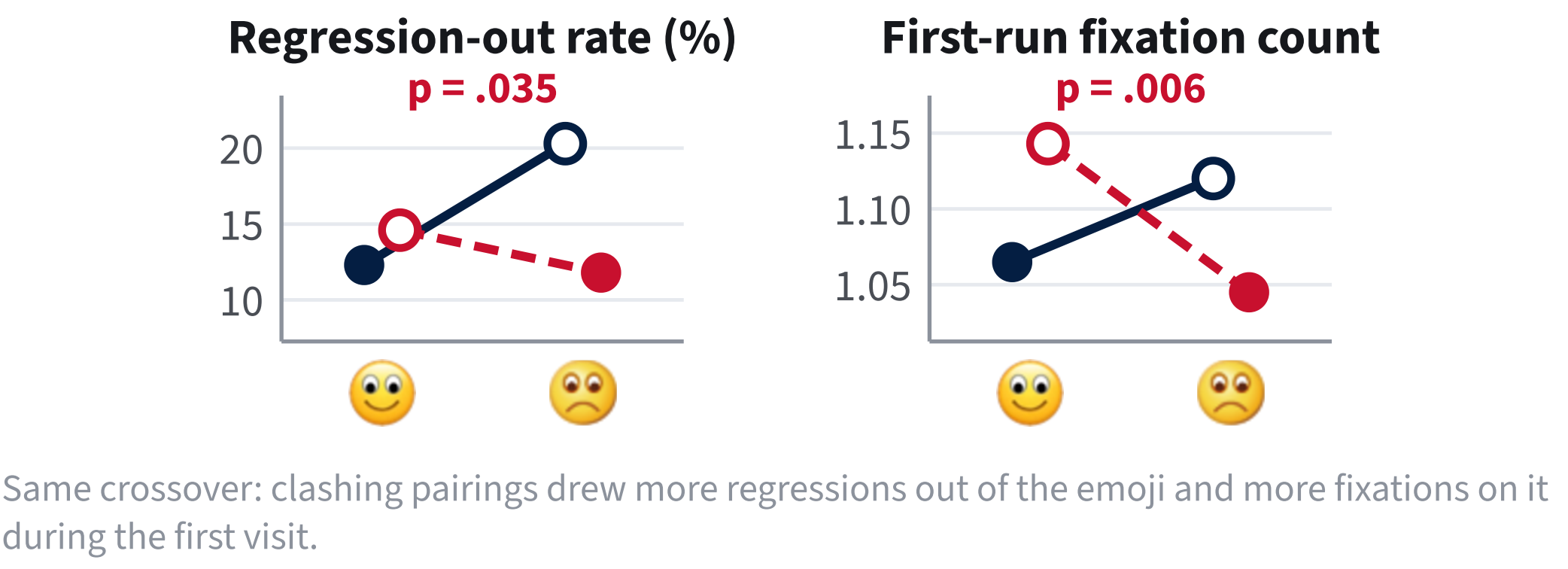
The clash shows up **once the eye lands on the emoji**. Readers then **go back to the evaluative word** to re-evaluate it.

SENTENCE TYPE × EMOJI VALENCE, AT THE EMOJI

Measure	Estimate	p
Go-past time	$\beta = -0.305$	.006
Total time	$\beta = -0.194$	.020
Regressions out	OR = 0.36	.035
First-run fixation count	$\beta = -0.101$	.006
Regressions back to the preceding word	OR = 0.28	.005

Negative β and OR < 1 both mean: the pairing that fits is read faster, with fewer looks back.

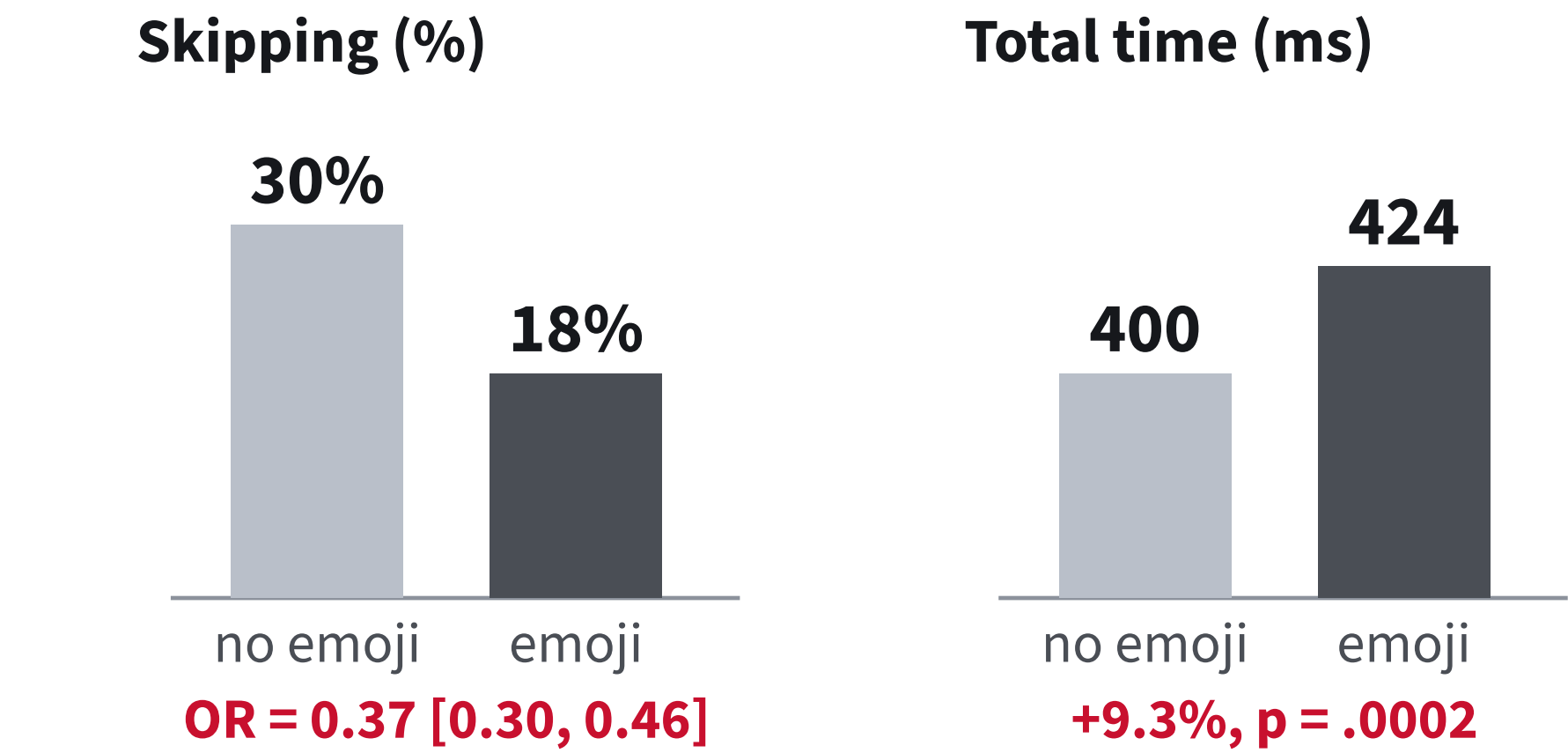
THE TWO EXPLORATORY MEASURES, ON THEIR OWN SCALES



Same crossover: clashing pairings drew more regressions out of the emoji and more fixations on it during the first visit.

2 · On the word before the emoji, presence shows up early and fit shows up late

Emoji presence changes how the preceding word is read.

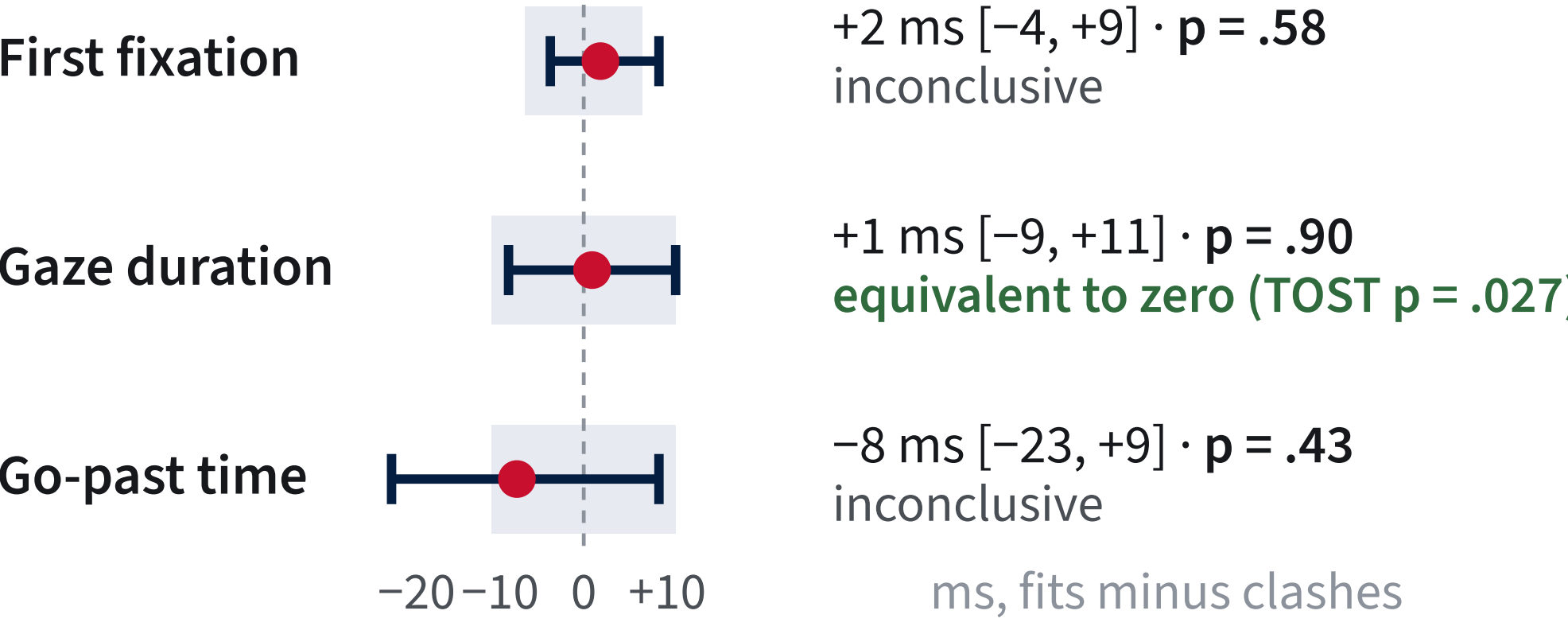


On the evaluative word, both sentence types behave alike. Total-time axis starts at 350 ms.

- Skipping falls from **30%** to **18%** when an emoji follows.
- Total time on the word rises by **9.3%**.

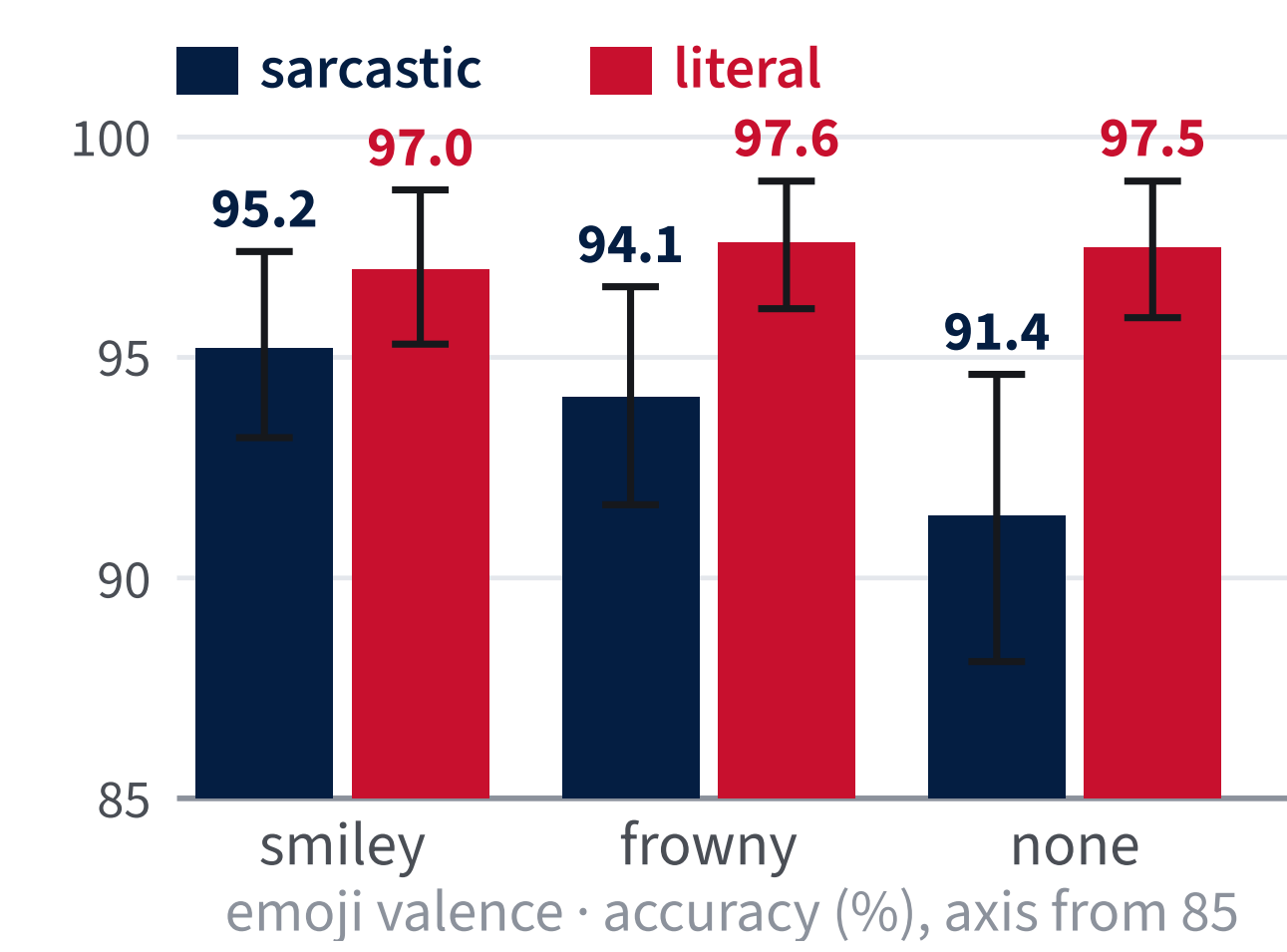
First-pass reading of that word is the same whether the emoji fits or clashes (all $p > .19$).

shaded band = the size of the object-emoji effect (Sheridan et al., 2024)



Estimates with 90% equivalence intervals, back-transformed from log units.

Comprehension accuracy by condition (overall 95.5%).



By-participant means with 95% CIs. Sarcastic sentences were understood less accurately than literal ones, 94% against 97% ($p < .001$). The lowest cell is sarcastic with no emoji; either emoji raises sarcastic accuracy by about 3 points ($p = .20$).

The eye picks up the emoji early and its fit later. Readers notice the emoji itself while it is still in the parafovea, and they notice how well it fits the word once the eye lands on it. For gaze duration the data are equivalent to zero against the object-emoji benchmark; first fixation and go-past remain inconclusive.

What this shows

- **Evaluative congruency behaves differently** from the referential word–picture match that produces early effects: emoji presence reliably changed skipping and later reading, while the early measures showed no evidence of it.
- The **congruency** between emoji and word shows up only once the eye lands on the emoji. Readers notice the clash there and **look back at the evaluative word** to re-evaluate it.

Open question

- Sarcastic items carry positive words and literal items negative ones, so **surface valence matching** and **sarcasm-marker fit** predict the same crossover.
- Separating them needs a fourth cell: **sincere praise**, where a smiley matches the word while the pragmatic frame differs.

Details

Sheridan, H., Barach, E., Christofalos, A. L., & Feldman, L. B. (2024). Emojis elicit semantic parafoveal-on-foveal effects during reading. *Visual Cognition*, 32(2), 151–161.

Huang, J., et al. (2025). Emoji position, eye movements and Chinese sarcasm comprehension. *Ergonomics*, 68(2), 251–266. Items adapted from this study.

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